



INTE130 – Object-Oriented Programming Assignment.
(CyberShield system Project)

Submitted Parts: Part C

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Introduction:

This design is based on the CyberShield idea from part B. CyberShield is a small cybersecurity company managed by one founder with the help of digital agents. The system helps small companies improve their cybersecurity and protect their systems from threats.

The digital agents support the founder by checking the system, finding problems and giving suggestions to improve security.

System Structure:

The CyberShield system works using a group of digital agents connected together. The founder is responsible for creating tasks and making the final decision.

When the founder creates a task the task is sent to the agents. Each agent does a specific job and returns results or recommendation. After reviewing all results, the founder decides what action should be taken.

The system works in this order:

Founder create a task.

Task is sent to agents.

Each the process the task.

Agents return results.

Founder reviews the results.

Founder makes the final decision.

This system helps organize the work and makes management easier for one person.

Multi agent structer:

The CyberShield system contains three main agents.

-Threat Monitring agent:

Responsiobility:

Monitor the system and detect suspicious activities.

Tasks:

Detect unusual loin attempts.

Monitor suspicious activites.

Find possible threats.

Information processed:

Login information.

User activity.

Security alerts.

-Vulnerability analysis agent

Responsibility:

Find weaknesses in the system.

Tasks:

Check weak passwords.

Detect old settings.

Scan system weaknesses.

Information Processed:

Password data.

Security settings.

System status.

Response agent:

Responsibility:

Give suggestions and security solution.

Tasks:

Suggest password changes.

Block suspicious access.

Give security recommendations.

Information processed:

Threat reports

Vulnerability results.

Security information.

System Classes:

The CyberShield system contains several connected classes.

Founder class”:

This class represents the founder of the company. The founder creates tasks , reviews the report, and makes the decisions.

Function:

Create_task()

Decision()

Task Class:

This class stores the task description that will be processed by the agents.

Attribute :

Description.

Threat agent class:

This class monitors threats and suspicious activities.

Function:

Process (task)

Vulnerability agent class:

This class checks vulnerability and weaknesses in the system.

Function:

Process(task)

Response Agent class :

This class gives recommendations and possible solution.

Function:

Process(task)

Relationship Between classes:

All classes in the CyberShield system are connected together.

-founder create the task.

-task stores the task information.

Task is sent to the task agent.

Task is sent to response agent.

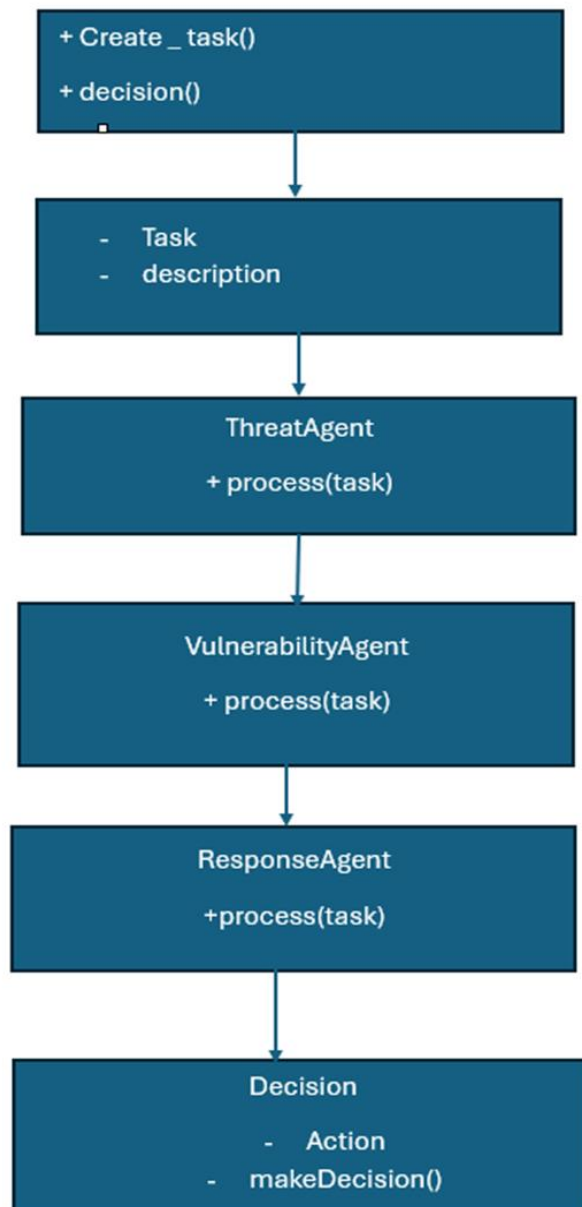
Agents process the task and return results.

Founder reviews the results.

Founder makes the final decision.

This connection helps the system work smoothly and clearly.

UML diagram:



UML diagram Explanation:

The UML diagram explains how the CyberShield system works.

Founder class is connected to task class.

Task is shared with all agents.

Each agent has a process(task) function.

Agent help the Founder by processing tasks.

Founder reviews the results and makes the final decision.

The UML diagram shows how all parts of the system are connected together.

Conclusion:

The Cybershield design explains how one person can manage a cybersecurity company using digital agents. The system is simple and organized and connected. Each part of the system has a clear role which helps the founder manage tasks more easily.